

Stay updated
with the latest
products in:

Surface Mount
Technology

LED

Electronics
Components

Power
Electronics

Test &
Measurement

Security &
Surveillance

Solar

Read
Electronics Bazaar

or Visit
www.electronicsb2b.com

delays imperceptible to the human brain. Naturally, apps would evolve amazingly.

Videos would be captured and transmitted on-the-go. Movies would be downloaded in seconds. You would be able to watch all your favourite television programmes on the phone itself. Social media would morph into a more exciting *avatar*, and so on. In a 2014 story by The Telegraph, the UK, experts predict that videos will represent 79 per cent of the data passing over the network by 2030. Standard-definition videos will disappear, and high-definition will become the norm, as 4K and 8K videos become more widespread. The download speed for a video stream is going to reach about 18Mbps, while the upload speed will also become increasingly important due to the popularity of apps like Instagram and iCloud. Sixty eight per cent of non-video demand will be made up by either augmented reality or mobile gaming.

That said, the real impact of 5G might be felt not just in entertainment and social apps, but also in businesses and other sectors like healthcare and automotive. Road-safety apps, for example, would become highly reliable as 5G would enable real-time, low-latency car-to-car communications. So, you would be getting instantaneous reports on the traffic situation and be alerted if a driver in front of you swerves suddenly.

Likewise, remote surgeries would improve with low-latency networks like 5G, because the doctor's reactions to the patient's condition would be captured immediately, just like in a real surgery.

The phone in your pocket

If the network is going to change, and also the apps, then the phone has to, obviously, change.

Wow, what clarity. If you are going to thrive on videos, surely you would need a better display on your mobile device. Several innovations, which seem cutting-edge today, such as automatic brightness adjustment,

extremely high resolutions, smooth transition from one environment to another and correction of defective vision, could become mainstream in the phones of 2020.

Capture it live. Problems like slow zoom-in, late focus and bad image quality in poor lighting conditions will not be acceptable to next-gen users. Going by the research happening around the world, we are likely to get smartphone cameras that will behave almost like our eyes, capable of focusing in real-time, tracking objects and automatically adjusting to the environmental lighting.

Less charging, more working. As mobile devices become such an integral part of our lives, we need to start worrying about not just data costs, but also power. Fortunately, researchers are working on super battery technologies, which will enable devices to work for longer with shorter charges. A team at Stanford University announced last year that they had designed a way of increasing the capacity of existing battery technology by 400 per cent.

Likewise, battery packs and power bank companies are innovating to provide a full charge to batteries within a few minutes, rather than taking almost an hour. If the trend continues, we should be able to run the dream phone of 2020 for a week or more with a few minutes' charge.

Supercomputing phones. Good network, great camera, super display—nothing would help if the processor is not up to it. I guess we need not worry on that front, with the low-power processor industry heating up with competition. If AT&T's 2012 report is to be believed, we are sure to have a supercomputing phone in our hands by 2020!

So, whether South Korea wins the race by trialling 5G at the 2018 Winter Olympic Games or Japan keeps its promise of launching a 5G trial network for the Summer Olympic Games in 2020 in Tokyo, one thing is for sure—there is a lot of excitement in store for us in the days to come. ●